



- 2 One end of a light elastic string of natural length a and modulus of elasticity $2mg$ is attached to a fixed point A on a rough horizontal surface. The other end of the string is attached to a particle P of mass m . The particle and string rest on the surface. The coefficient of friction between P and the surface is μ . The particle P is initially held in equilibrium at a distance $\frac{4}{3}a$ from A . The particle is then released from rest.

(a) Given that the string never becomes slack, find the minimum value of μ . [3]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.



(b) Find the extension of the string when the particle comes to rest.

[3]

This image shows a full page of a document template designed for handwriting practice or general writing. It consists of approximately 20 evenly spaced, horizontal dotted lines extending across the entire width of the page. The background is plain white, and there are no margins, headers, footers, or other markings present.